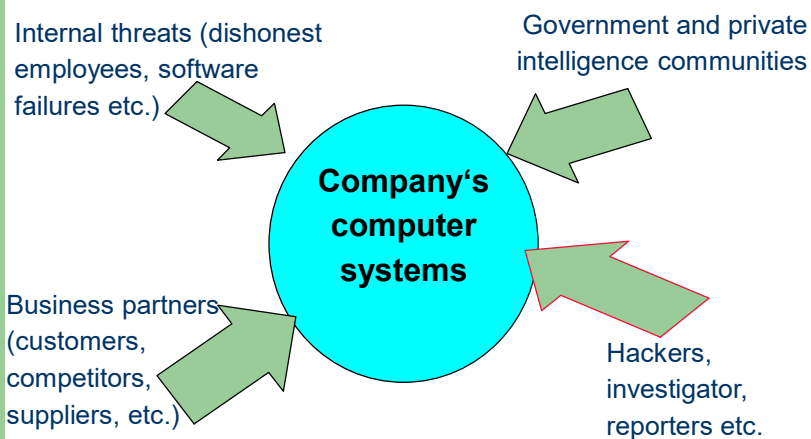
Background

- Information Security requirements have changed in recent times
- traditionally provided by physical and administrative mechanisms
- computer use requires automated tools to protect files and other stored information
- use of networks and communications links requires measures to protect data during transmission

9

Computer Security and Industries



10

Security issues

- Robbery through computers
- Identity theft
- Loss of privacy
- Industrial espionage on corporate information
- Information vandalism
- Computer viruses/ malware

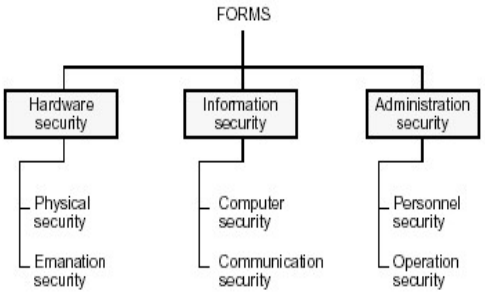
11

Computer Security

- Attacks can be INTERNAL and EXTERNAL.
- INTERNAL:
  - altering data;
  - stealing source code;
  - damaging computer systems;
  - revealing confidential information;
  - intentionally writing bad code for later use;
  - etc.
- EXTERNAL
  - Send malicious programs;
  - Scanning your network for vulnerabilities and attack it;
  - Sending viruses/ worms
  - Etc.

12

Forms of Security



13

Types of Security

- Information Systems Security: protect system.
  - hardware and/or software functions
  - operational and accountability procedures and management
  - central and remote access controls
  - physical structures/ devices

14

Types of Security, cont'd.

- Information Security: protect data
- Network Security: protect network
  - Secure computers from denial of service and unauthorized access/ use.
- Software/ Application Security: protect software applications
  - Ensure software and applications are reliable
- Computer Security: protect all computing resources

15

What is Security ?

16

What is Security ?

Often defined by six properties

- Also called services, capabilities, goals

17

Six Properties

1. Confidentiality
2. Integrity
3. Availability

18

Security Properties, cont'd.

4. Authentication
5. Authorisation
6. Non-repudiation

19

Confidentiality

- Data can be sent without revealing contents
- The property that information is not made available or disclosed to unauthorised individuals, entities or processes.

20

Integrity

- Data Integrity: the property that data has not been altered or destroyed in an unauthorised manner.
  - data can be sent accurately
- System Integrity : the requirement that all system assets are operating according to specification and in the way that the current user believes them to be operating.
  - service performs accurately

21

Availability

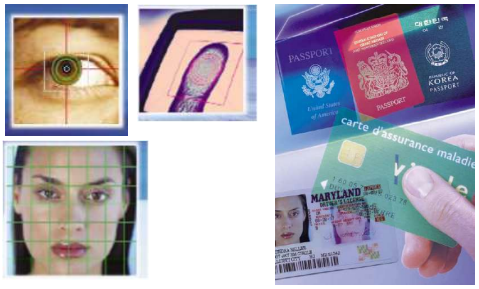
- a service/ data is available for use when needed



22

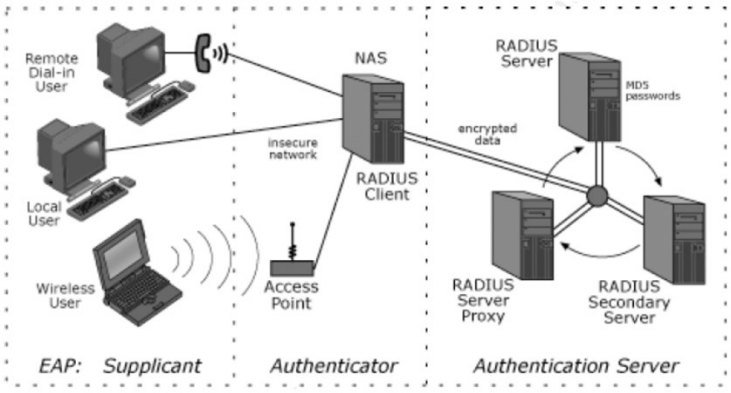
Authentication

- the identification of users and/ or services can be verified
- The verification of a claimed identity.



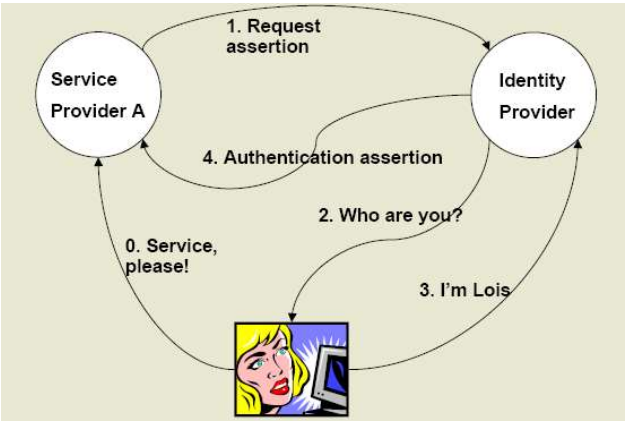
23

Authentication



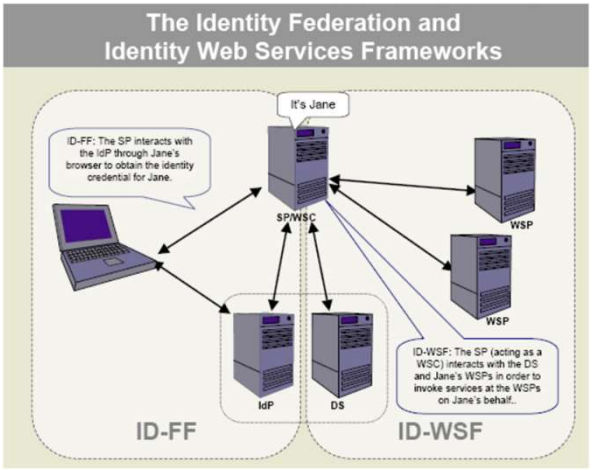
24

Authentication



25

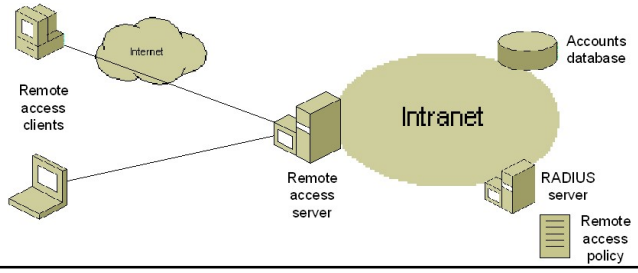
Authentication



26

Authorisation

- users and/or services can be allowed to only perform certain operations
- The process of granting or denying access to a network resource.



27

Two Step Process

Most computer security systems are based on a two-step process.

- The first stage is authentication, which ensures that a user is who he or she claims to be.
- The second stage is authorization, which allows the user access to various resources based on the user's identity.

28

Non-Repudiation

- if a user is authenticated and authorised, they can not retract their transactions
- To ensure that a transferred message has been sent and received by the parties claiming to have sent and received the message.
- Non-repudiation is a way to guarantee that the sender of a message cannot later deny having sent the message and that the recipient cannot deny having received the message.

29

Aspects of Security

- Security attack:
  - Any action that compromises the security of information owned by an organization.
- Security mechanism:
  - A process (or a device incorporating such a process) that is designed to detect, prevent, or recover from a security attack.
- Security service:
  - A processing or communication service that enhances the security of the data processing systems and the information transfers of an organization. The services are intended to counter security attacks, and they make use of one or more security mechanisms to provide the service.

30

?



31

OSI Security Architecture

- ITU-T X.800 “Security Architecture for OSI”
- defines a systematic way of defining and providing security requirements

32

Terminology

- threat
  - circumstances that have the potential to cause loss or harm
- vulnerability
  - a weakness in a computer system that might be exploited to cause loss or harm
- attack
  - an action that exploits a vulnerability
- control
  - a protective measure

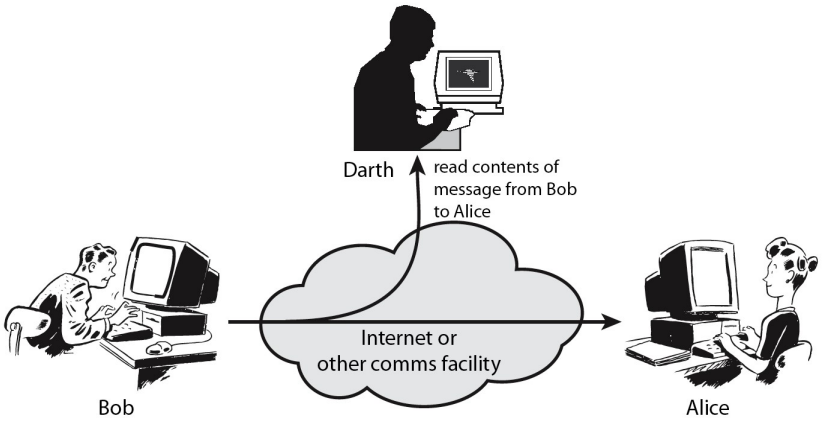
33

Security Attack

- any action that compromises the security of information owned by an organization
- information security is about how to prevent attacks, or failing that, to detect attacks on information-based systems
- often *threat* & *attack* used to mean same thing
- have a wide range of attacks
- can focus of generic types of attacks
  - passive
  - active

34

Passive Attacks



35

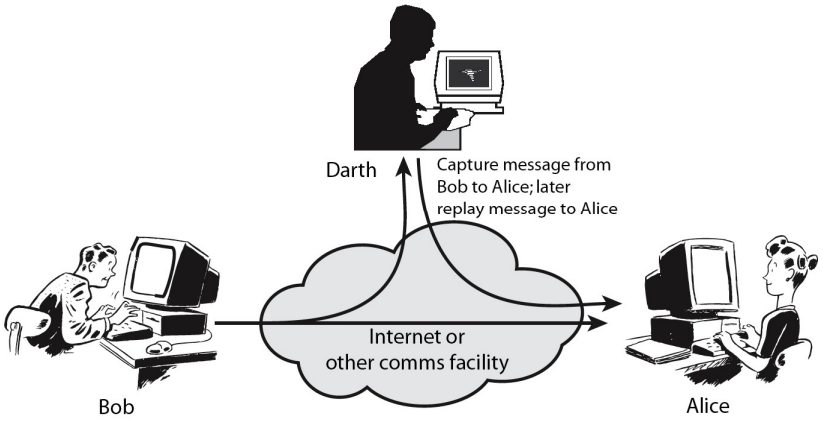
Attacks – Passive Types

Passive (***interception***) – eavesdropping on, monitoring of, transmissions.

- The goal is to obtain information that is being transmitted.
- Types here are: release of message contents and traffic analysis.

36

Active Attacks



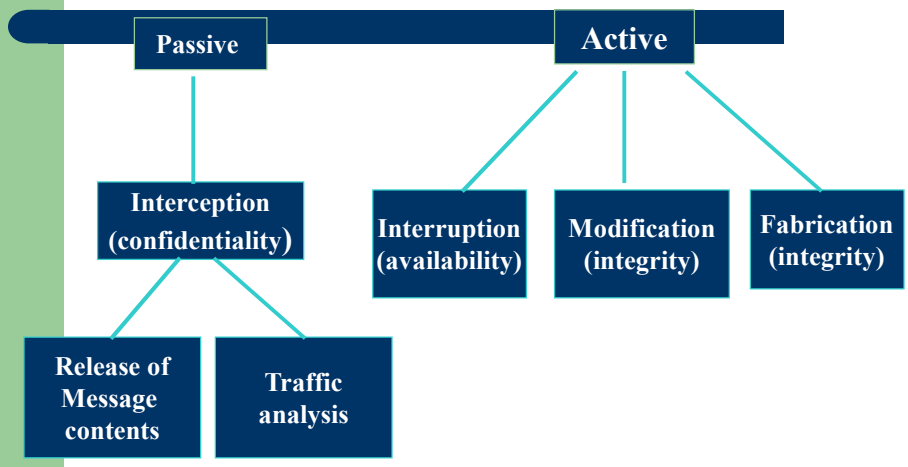
37

Attacks – Active Types

- Involve modification of the data stream or creation of a false stream.
- It can be subdivided into –
  - masquerade
  - modification of messages
  - denial of service

38

Attacks



39

Internet Attacks

Publicly accessible networks are at risk from opportunistic attackers using network scanning tools to find and compromise vulnerable hosts.

There are two groups:

- Those who compromise hosts for denial-of-service and flooding purposes
- Those who compromise hosts through which attacks can be bounced (including port scans, breaking into other hosts, or sending spam email)

The second type of threat is posed by determined attackers who probe every point of entry into a target network from the Internet.

40